Namrata Srivastava, PhD

Research Scientist | Learning Analytics | HCI | AI in Education
Nashville, TN | namratasrivastava.research@gmail.com
namratasri.com | linkedin.com/in/namratasri01 | Google Scholar

SUMMARY

Research scientist with 8+ years of experience measuring how people learn with technology, across learning analytics, HCI, and AI. I design experiments and analytics that go beyond engagement metrics to capture learning itself: students' strategies, inquiry processes, attention, and cognitive load, using data ranging from eye-trackers to 7M+ institutional records. Most recently, I built and evaluated LLM-powered agents and teacher dashboards in K–12 classrooms, and I regularly translate findings for teachers, policy partners, and public audiences. Published at AAAI, CHI, LAK, and AIED; 820+ citations, h-index 18.

SKILLS

Data Science: Python (pandas, scikit-learn), R, SQL, data pipelines, statistical modeling, hypothesis testing, predictive modeling

Measuring Learning: Learner modeling, sequence and process mining, Markov models, self-regulated learning analytics, engagement and affect detectors

Research Methods: Experiment design, classroom field studies, controlled user studies, surveys and interviews, thematic analysis

AI and LLMs: GPT-4o, Llama-3.1-8B (locally hosted), retrieval-augmented generation (RAG), prompt engineering, rubric-based LLM evaluation

Sensing and Tools: Eye-tracking, thermal imaging, physiological sensing, multimodal data (audio, video, logs), Flask, HTML/JavaScript, R Shiny, Git

EXPERIENCE

Research Scientist **Jun 2024 – Present**

Vanderbilt University, Nashville, TN

- Led design and classroom evaluation of **SciStory**, a game-based science environment with role-specific LLM agents; ran a study with **78 sixth graders** (GPT-4o) and analyzed a 15-pair pilot (locally hosted Llama-3.1-8B). Markov models and sequence mining of game logs showed each agent drove a distinct phase of scientific inquiry; **82%** of the 73 students surveyed said the agents prompted them to revise their writing (co-first author, LAK 2026; AIED 2026 demo; ISLS 2026).
- Run end-to-end data pipelines for classroom studies, using SyncFlow to capture synchronized audio, video, and game logs from student pairs, then processing and aligning the streams for analysis. Currently analyzing SciStory's multimodal data to understand how students take up AI feedback and improve their arguments over time.
- Built **LearnLens**, a generative-AI teacher dashboard that summarizes students' open-ended responses and flags common misconceptions. Iterated prompts with human validation to reduce hallucinations; interviews with sixth-grade teachers showed they trusted data-grounded summaries but not prescriptive advice, shaping how AI insights are presented.
- Collaborated on classroom eye-tracking research with a PhD student, including using LLMs to turn multimodal reading traces into assessment reports for teachers (AIED 2025) and recognizing students' in-the-wild reading behaviors (ECCV 2026).
- Co-designed STEM+C curriculum and classroom tools (inquiry interfaces, a block-based coding environment, teacher guides) with teachers through professional development, working across learning scientists, engineers, and schools (NSF SPICE).
- Mentor 2 PhD students, including one building AI agents for 4th–5th grade math with the OECD; contribute to federal grant proposals as PI, Co-PI, and senior personnel.

Postdoctoral Researcher **Nov 2023 – May 2024**

Penn Center for Learning Analytics, University of Pennsylvania, Philadelphia, PA

- Built at-risk student prediction models on a large MOOC dataset using SQL and Python, and audited model fairness across gender and demographic groups.

- Mentored PhD students developing detectors of boredom, confusion, and carelessness from intelligent tutoring system logs.

Research Fellow **Sep 2020 – Oct 2023**

Centre for Learning Analytics, Monash University, Melbourne, Australia

- Led curriculum analytics on **7M+ course enrollment records** (SQL, R, Python), linking course choices and sequencing to academic performance; co-designed a dashboard for instructors and administrators to identify bottleneck courses. **The Faculty of IT adopted the recommendations** (first author, JLA 2024).
- Identified self-regulated learning strategies from the trace data of **253 students** using sequence and process mining (first author, LAK 2022).
- Supervised development of an early-warning system for at-risk students with targeted feedback interventions (JLA), and co-supervised a PhD on fairness and trust in educational predictive analytics (BJET 2025, LAK 2024).

Machine Learning Research Intern and Research Collaborator **Jul 2020 – 2025**

Adobe Research, Remote

- Built a mobile web app (HTML/JavaScript) that delivers AI-generated reading primes as text summaries (GPT-4), images (Midjourney V5), and mind maps; evaluated them with mobile readers in user studies (CHI 2025 Extended Abstracts).
- Designed review-and-preview support to reduce the cost of reading interruptions (CHI 2021 Extended Abstracts), and mentored a Master's student predicting deep vs. skim reading from eye-tracking using gaze features I developed (CHI 2023).

EDUCATION

Ph.D., Computer Science **2016 – 2021**

University of Melbourne, Australia

- Thesis: Towards sensor-based learning analytics. Built a contactless eye-tracking and thermal-imaging system to measure students' cognitive load, and ran user studies with 100 participants (published at CHI, LAK, and IMWUT).

M.Tech., Information Technology **2013 – 2015**

Indian Institute of Information Technology, Allahabad, India

AWARDS AND HONORS

- Judge's Choice Award, EdTechnical AIEd Forecasting Competition (2026)
- LIVE Ignite Tech Development Grant, Vanderbilt University, for an AI ethics chatbot (2025 – 2026)
- Featured among trailblazing women researchers in STEM, Herald Sun (2025)
- Emerging Scholar Award, Society for Learning Analytics Research (2024)

LEADERSHIP AND SERVICE

- Executive Committee (Secretary), Society for Learning Analytics Research (2024 – 2026)
- Proceedings Chair, AIED 2026; lead organizer of the CROSSMMLA workshop at LAK and AIED for the past four years

TEACHING AND COMMUNICATION

- Instructor, undergraduate data science (DS 1000), Vanderbilt University (90 students, Fall 2026; 40 students, Spring 2026)
- Chief Examiner, postgraduate online data science course, Monash University; taught Python-based data processing, University of Melbourne
- Communicated research to non-technical audiences through invited talks and panels (Universidad Autónoma de Madrid, 2025; AI Scholar series, NC State University, upcoming), live system demos (AIED 2026), and public talks at university open days and exhibitions